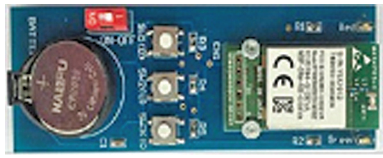
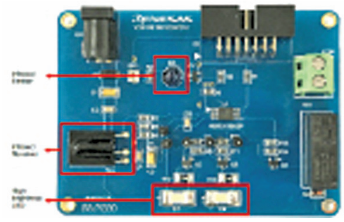
For Smart Homes/Offices

The author, Sharad Bhowmick, works as a Technology Journalist at EFY. He is passionate about power electronics and energy storage technologies. He wants to help achieve the goal of a carbon neutral world.

Wi-Fi Based Remote Control for Lighting and Other Appliances	PowerPoint Presenter	Infrared Human Sensor
		
This reference design is for a universal remote control that can be used to switch on/off any appliance or light from your mobile device. It can help you automate your house.	This reference design of a low-powered PowerPoint presenter enables users to control slides by exchanging data with a PC through Bluetooth Low Energy (BLE) communication. It has an operational range of up to 30 metres.	This reference design is for a low-cost human presence detector. The design can help you automate lighting, doors, and appliances to work automatically when someone walks in or walks out.
The reference design from Microchip lets you connect and program smart devices, smart appliances, and smart gadgets in your home. It employs a Curiosity PIC32MZ EF 2.0 development board featuring a 32-bit PIC32MZ EF microcontroller. The reference design can also be used to control the red/green/blue (RGB) LED lights from the smartphone or a PC. The reference design requires a WINC1500 Xplained Pro Extension Board plugged into the Curiosity board's XPRO connector to enable and configure Wi-Fi connectivity. A web page hosted by the Curiosity board provides the interface to control any appliance or change the colour of the onboard RGB LED.	This reference design from Renesas is a PowerPoint (PPT) presenter solution that enables the user to easily change the page by pressing a page-down key or page-up key on the PPT presenter when showing a PPT file. The presenter allows the user to wirelessly control and present the slides without any human intervention with the computer. The remote connects with the computer or multimedia projector via Bluetooth Low Energy (BLE). The reference design is based on the RL78/G1D module and works on a 3V CR2032 lithium battery and has a low power consumption of 2.6 mA in RF low power consumption mode. It has a working range of up to 30 metres and is compatible with both Windows and Android platforms.	This reference design from Renesas is a low-cost infrared human sensor that is based on the reflective infrared induction principle. The energy-saving design requires few components to construct, making it inexpensive and ideal for lighting entrances, porches, stairs, and bathrooms. The reference design can also be incorporated with the automatic doors, fans, and other appliances to automate them. The design uses an RL78/G10 microcontroller (MCU), which is a 10-pin LSSOP general-purpose MCU. When people come close, the bulb lights up, and when they leave, the bulb switches off automatically. This design can work with a 220V AC supply and has a low power consumption. It is capable of relaying an output of up to 3A/250V AC and has a fast response time.
The reference design is suitable for use in residential and commercial purposes.	The reference design can be used for presenting slides during meetings and presentations.	The reference design can be used in residential or commercial buildings, industrial setups, and even in public places to detect human presence.
Microchip	Renesas	Renesas
https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-wi-fi-based-remote-control-for-lighting-and-other-appliances	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-of-a-powerpoint-presenter	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/this-reference-design-can-be-integrated-with-light-to-build-temporary-lighting-in-entrances-porches-stairs-and-bathrooms